Elizabeth H. Childs

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Mechanical engineer and HCI researcher who takes tangible, haptic, and AR interfaces from concept through hardware/software prototype to evidence-based user evaluation.

Education

Stanford University *Stanford, CA*
PhD, Mechanical Engineering (Human-Computer Interaction Research), GPA 3.9/4.0 *Expected Dec 2027*
MS, Mechanical Engineering (Controls); MA, Education *May 2026*
Knight-Hennessy Scholar: Stanford's flagship graduate leadership fellowship (<1% acceptance rate)
Impact Labs PhD Fellow: competitive fellowship for solutions-oriented research with practitioners

University of Maryland, College Park *College Park, MD*
B.S., Mechanical Engineering, GPA 4.0/4.0 *Dec 2020*
Banneker/Key Scholar: UMD's most selective merit scholarship (full tuition, room, and board)

Research & Industry Experience

Stanford University, Stanford HCI Group & CHARM Lab *Stanford, CA*
PhD Researcher (Advisors: James Landay, Allison Okamura) *June 2021 – Present*
- Led a program on tangible and AR interfaces for hands-on learning, owning projects from needfinding and concept through prototyping, user studies, and publication.
- Led needfinding across 7 schools (58 teacher surveys, 18 semi-structured interviews, classroom observation); findings set research direction to augment physical labs rather than replace them.
- Built an AR physics lab over a real pendulum; in a 52-student between-subjects study it improved scientific reasoning vs. a traditional lab ($d = 0.63$).
- Built an AR biology app (Unity, OpenCV) that tracks students' real balloons and animates the enzyme activity behind them in sync with their data; deployed at 5 institutions with 100+ students.
- Built the AR features for a 10-person team's narrative outdoor-learning app, evaluated in a field study with 44 children (ACM CHI 2023).

Microsoft Research, Future Experiences Group *Redmond, WA*
Research Intern (Mentors: Sumit Basu, Scott Saponas) *June 2026 – Sep 2026*
- Conceived, designed, and built a free-moving haptic robot that provides multidirectional forces and torques so learners can feel physics simulations, from airfoil lift to molecular bonds.
- Engineered a gearless omni-wheel drivetrain with custom motor control, packaging a Raspberry Pi, onboard camera, battery, and buck/boost power electronics into one compact 3D-printed body.
- Developed hybrid localization fusing overhead computer-vision tracking with motor feedback, keeping force rendering precise between camera frames; integrated with projected-AR physics simulations.

Dolby Laboratories, Advanced Technology Group *Sunnyvale, CA*
Research Intern *May 2024 – Aug 2024*
- Conceived and prototyped a hybrid microscope whose physical focus, magnification, and stage controls drive high-resolution imagery on a connected smartphone; presented the concept to ATG leadership.

Lam Research, Global Products Engineering *Fremont, CA*
Mechanical Engineering Intern *May 2021 – Aug 2021*
- Developed HoloLens AR applications (Unity, C#) that render a live hologram of industrial robots running inside opaque enclosures, turning telemetry into spatial views for engineers.
- Self-initiated a tool that replays robot motion leading up to a fault, helping technicians better diagnose errors and demonstrating how AR can improve robot servicing.

University of Maryland, GAMMA Lab *College Park, MD*
Research Assistant (Advisor: Dinesh Manocha) *Aug 2020 – Aug 2021*
- Co-led (co-first author) an IEEE TVCG survey of how AR/VR can support distance learning.

Mechanical Engineering & Robotics Experience

University of Maryland, Bioinspired Advanced Manufacturing Lab *College Park, MD*
Undergraduate Researcher (Advisor: Ryan Sochol) *Sept 2018 – July 2020*

- Invented an additive assembly method for multi-material 3D-printed microfluidics (open halves with printed fasteners and seals), cutting post-processing time ~98% at equal or higher burst pressure.
- Led a 3-person team (1 undergraduate, 1 master's student) through first-author publication.

University of Maryland, Bioinspired Robotics: Flic-Flac Spider Robot *Feb 2019 – May 2019*

- Designed and built a somersaulting robot inspired by the flic-flac spider using 4 coordinated servos, a Teensy, and IMU orientation sensing; it covered the course 20× faster than required, fastest of 13 teams.

Key Technologies, Inc. *Baltimore, MD*
Mechanical Engineering Intern, Medical Device Consulting *Sept 2019 – Dec 2019*

- Built medical device prototypes and designed and fabricated test rigs for microfluidic devices.

Institute of Technology of Cambodia *Phnom Penh, Cambodia*
NSF International Research Experience for Students (Advisor: Garrett Clayton) *June 2019 – Aug 2019*

- Designed a compact modular robot to reach unexploded ordnance in dense forest.

Oregon State University, Dynamic Robotics Lab *Corvallis, OR*
NSF REU Researcher (Advisor: Jonathan Hurst) *June 2018 – Aug 2018*

- Designed impact-absorbing feet for the Cassie bipedal robot, force-plate testing 3 geometries in silicones, foams, and gels; cut peak force 47% and step impact 90% with no controller changes.

Selected Publications

1. **E. Childs***, F. Mohammad*, L. Stevens* et al., "Enhancing Distance Learning Through AR and VR Technologies," *IEEE TVCG*, 2024.
2. A. Cheng, . . . , **E. Childs**, et al., "Designing Immersive, Narrative-Based Interfaces to Guide Outdoor Learning," *ACM CHI*, 2023.
3. U. Bhattacharya, **E. Childs**, et al., "Speech2AffectiveGestures: Synthesizing Co-Speech Gestures with Generative Adversarial Affective Expression Learning," *ACM Multimedia*, 2021.
4. **E. Childs** et al., "Additive Assembly for PolyJet-Based Multi-Material 3D Printed Microfluidics," *J. MEMS*, 2020.

Talks, Panels & Outreach

Top Poster & Talk, poster selected for a presentation, Augmented World Expo, Long Beach 2026
Panelist, SERGE outreach panel for ~20 students 2026
Workshop Co-Organizer, Purposeful XR: Ethical Futures, ACM CHI, Yokohama 2025
Panelist, The Educational Rift in Spatial Reasoning, Augmented World Expo, Long Beach 2024
Panel Moderator & Demo Presenter, XR in Education, Stanford XR Conference 2022–23

Honors & Awards

1st Place, Best Use of Looking Glass, MIT Reality Hack (90+ teams) 2024
Academic Achievement Award, highest GPA of ~300 UMD mechanical engineers 2019
NSF LSAMP Bridge Scholar, selective NSF-funded program for incoming UMD students 2016

Skills

Research Methods: needfinding, interviews, surveys, qualitative coding, experiments, Bayesian analysis
Hardware: SolidWorks, 3D printing (FDM, PolyJet), CNC machining, laser cutting, PDMS (silicone) manufacturing, mechatronics, motor control, power electronics, sensor fusion, embedded systems
Software & XR: Unity, C#, C++, computer vision (OpenCV), ML, HoloLens, AR/VR, Maya